

(a) carbonyl migration removed					(b) dehydration to the enol removed					(c) Cannizzaro reaction removed				
Threose	1→1	1→1	3→3	3→3	Threose	1→1	1→1	3→lost	3→lost	Threose	1→1	1→1	3→3	3→3
Ribose	2→2	2→2	11→13	—	Ribose	2→2	2→2	11→lost	—	Ribose	2→2	2→2	11→11	—
Glycerol	2→2	2→2	—	—	Glycerol	2→2	2→2	—	—	Glycerol	2→lost	2→lost	—	—
Deoxyribose	3→3	3→3	11→13	—	Deoxyribose	3→lost	3→4	11→lost	—	Deoxyribose	3→5	3→5	11→11	—
Glyoxylate	6→6	3→3	11→11	5→5	Glyoxylate	6→lost	3→3	11→lost	5→lost	Glyoxylate	6→6	3→lost	11→12	5→lost
	F	FA	G	GA		F	FA	G	GA		F	FA	G	GA

reached in the same number of steps
 still reached, but by a longer route
 no longer reachable
 not reachable with the whole rule set